

OCR Further Mathematics
Mechanics - Further Dynamics
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
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Question			Answer/Indicative content	Marks	Part marks and guidance	
1	a		$F \propto \frac{1}{(t+1)^2}$ $\therefore F = \frac{k}{(t+1)^2} = ma = 3 \frac{dv}{dt} \Rightarrow \frac{dv}{dt} = \frac{k}{3(t+1)^2}$	B1 (AO3.1b) [1]	AG	
	b		$\therefore v = \frac{k}{3} \int \frac{1}{(1+t)^2} dx = \frac{-k}{3(1+t)} + u$ $t = 0, v = 0 \Rightarrow k = 3u$ $t = 1, v = 2 \Rightarrow 2 = \frac{-k}{3(1+1)} + u$ $\Rightarrow u = 4, k = 12 \Rightarrow v = 4 - \frac{4}{1+t}$ oe	M1 (AO3.1b) M1 (AO3.1b) M1 (AO3.1b) A1 (AO1.1) [4]	Separating variables correctly and integrating to $\frac{C}{1+t}$; award if “+ u” missing kg Substituting initial values to determine a relationship between k and u . Substituting $t = 1$ to determine a second relationship between k and u oe. eg $v = \frac{4t}{1+t}$	May use + c instead of u NB The units of k are Ns^2 or kg m but these are not required.

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Question			Answer/Indicative content	Marks	Part marks and guidance		
	c		$\frac{dx}{dt} = 4 - \frac{4}{1+t} \Rightarrow x = 4t - 4\ln(1+t) + c$ $t = 0, x = 1 \Rightarrow c = 1 \text{ so } x = 4t - 4\ln(1+t) + 1$	M1 (AO1.1) A1 (AO1.1) [2]	For integrating their 'v' to reach an expression involving $k \ln(1+t)$ oe Can be awarded even if no "+ c"		
	d		95% of $v_T = 0.95 \times 4 = 3.8$ $v = 3.8 \Rightarrow 3.8 = 4 - \frac{4}{1+t}$ $\Rightarrow 0.2 = \frac{4}{1+t} \Rightarrow 1+t = 20 \Rightarrow t = 19$ so $x = 4 \times 19 - 4\ln(1+19) + 1$ $x = 77 - 4 \ln 20 \text{ so distance moved is } 76 - 4 \ln 20 \text{ m or awrt } 64 \text{ m}$	B1 (AO2.2a) M1 (AO3.1b) A1 (AO1.1) M1 (AO1.1) A1 (AO1.1) [5]	Setting their v to their 3.8 in the appropriate equation Substituting their t into the equation for x		
			Total	12			



Question			Answer/Indicative content	Marks	Part marks and guidance	
2	a		[Force] = MLT^{-2} [Viscosity] = $MLT^{-2}L^{-2}T = ML^{-1}T^{-1}$	M1 (AO1.2) A1 (AO1.1) [2]		
	b		No it is not possible since a multiplicative constant does not affect the dimensions of the quantity it is multiplying.	B1 (AO2.4) [1]	Or equivalent statement about dimensional analysis only applying to dimensioned quantities	
	c		$MLT^{-2} = (ML^{-1}T^{-1})^\alpha L^\beta (LT^{-1})^\gamma$ From M, $\alpha = 1$ L: $1 = \beta + \gamma - \alpha$, T: $-2 = -\alpha - \gamma$ $\beta = 1, \gamma = 1$	M1 (AO3.3) B1 (AO1.1) M1 (AO1.1) A1 (AO1.1) [4]	Setting up a dimensional equation (using their [Force] and [Viscosity]) Using their equation to derive 2 equations in β and γ	$MLT^{-2} = M^\alpha L^{\beta+\gamma-\alpha} T^{-\alpha-\gamma}$ NB – B1 to allow this from M0

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Question		Answer/Indicative content	Marks	Part marks and guidance		
	d	$mg - 6\pi\eta rv = m \frac{dv}{dt}$ $\int dt = \int \frac{1}{g - kv} dv$ $t + c = -\frac{1}{k} \ln(g - kv)$ $t = 0, v = c = -\frac{1}{k} \ln g$ $0 \Rightarrow$ $t - \frac{1}{k} \ln g = -\frac{1}{k} \ln(g - kv) \Rightarrow -kt = \ln(g - kv) - \ln g$ $\Rightarrow -kt = \ln\left(\frac{g - kv}{g}\right) \Rightarrow e^{-kt} = \frac{g - kv}{g}$ Alternative method 1 $mg - 6\pi\eta rv = m \frac{dv}{dt}$ $g = kv + \frac{dv}{dt} \text{ so Integrating Factor}$ $e^{\int k dt} = e^{kt}$ $ge^{kt} = kve^{kt} + \frac{dv}{dt} e^{kt} = \frac{d}{dt}(ve^{kt})$ $ve^{kt} = \int ge^{kt} dt = \frac{g}{k} e^{kt} + c$ $t = 0, v = 0 \Rightarrow c = -\frac{g}{k}$ $ve^{kt} = \frac{g}{k} e^{kt} - \frac{g}{k} \Rightarrow e^{kt} \left(\frac{g}{k} - v\right) = \frac{g}{k}$ $\Rightarrow e^{-kt} = \frac{\frac{g}{k} - v}{\frac{g}{k}} = \frac{g - kv}{g}$ Alternative method 2 $mg - 6\pi\eta rv = m \frac{dv}{dt}$ $g = kv + \frac{dv}{dt}$ $C \quad \frac{dv}{dt} + kv = 0 \Rightarrow \frac{dv}{dt} = -kv \Rightarrow \int \frac{1}{v} dv = -\int k dt$ F:	M1 (AO3.3) A1 (AO1.1) M1 (AO1.1) M1 (AO3.4) A1 (AO1.1) M1 A1 M1 M1 A1 M1 A1 M1	Using $F = ma$ with two forces Correct Correctly separating variables in v and t Integrating and substituting initial values to find c AG. Must see some intermediate working Using $F = ma$ with two forces Rearrange and find IF Substituting initial values to find c AG. Must see some intermediate working	any differential expression for a no need for k to be substituted yet any differential expression for a no need for k to be substituted yet	



Question			Answer/Indicative content	Marks	Part marks and guidance		
			$\Rightarrow \ln v = -kt + c \Rightarrow v = Ae^{-kt}$ PI: $v = \alpha \Rightarrow k\alpha = g \Rightarrow \alpha = \frac{g}{k}$, so try $v = Ae^{-kt} + \frac{g}{k}$ GS: $t = v = 0 \Rightarrow 0 = A + \frac{g}{k} \Rightarrow A = -\frac{g}{k}$ 0, So $v = -\frac{g}{k}e^{-kt} + \frac{g}{k} \Rightarrow e^{-kt} = \frac{k}{g}\left(\frac{g}{k} - v\right) = \frac{g - kv}{g}$	M1 A1 [5]	Using $F = ma$ with two forces Rearrange and find CF and PI	any differential expression for a no need for k to be substituted yet	
e			$e^{-kt} = \frac{g - kv}{g}$ so $t \rightarrow \infty \Rightarrow v_T = \frac{g}{k}$ Alternative method $\frac{dv}{dt} = 0 \Rightarrow mg - 6\pi\eta r v_T = 0 \Rightarrow v_T = \frac{mg}{6\pi\eta r} = \frac{g}{k}$	B1 (AO3.4) B1 [1]			
f			As the viscosity increases the terminal velocity decreases... ...and as the viscosity tends to infinity the terminal velocity tends to 0	B1 (AO2.2a) B1 (AO2.2a) [2]			
			Total	15			



Question			Answer/Indicative content	Marks	Part marks and guidance		
3	a		30×10 300 J	M1 (AO1.1) A1 (AO1.1) [2]	Using Work done = Fd		
	b		$\frac{1}{2} \times 2.4 \times 12^2 + 300 = \frac{1}{2} \times 2.4 \times 18^2 + W$ $10R = 84$ $R = 8.4$ Alternative method $a = \frac{18^2 - 12^2}{2 \times 10} = 9$ $30 - R = 2.4 \times 9$ $R = 8.4$	M1 (AO1.1) M1 (AO1.1) A1 (AO1.1) M1 M1 A1 [3]	where W is energy loss (could be eg $10R$) Use of energy loss = $10R$ Using $v^2 = u^2 + 2as$ to find a Use of $F = ma$ with their a	Allow 1 slip e.g. R instead of W Must be e.g. $10R$, not R	
	c		$t = \frac{2 \times 10}{18 + 12}$ $t = \frac{2}{3}$ $\frac{84}{2/3} = 126 \text{ W}$ Alternative method $t = \frac{18 - 12}{9}$ $t = \frac{2}{3}$ $\frac{84}{2/3} = 126 \text{ W}$	M1 (AO1.1) A1 (AO1.1) A1 (AO1.1) M1 A1 A1 [3]	Using $s = \left(\frac{v+u}{2} \right) t$ Using $v = u + at$ to find t	Or use Average power = Force $\times$ average speed, i.e. from (b) or N2L:	
	d	i	$17.28 = 1.6 \times v_Q$ $v_Q = 10.8 \text{ m s}^{-1}$	M1 (AO1.1) A1 (AO1.1) [2]	Do not allow -10.8		



Question			Answer/Indicative content	Marks	Part marks and guidance		
		ii	$2.4 \times 18 = 2.4 \times v_P + 1.6 \times 10.8$ So $v_P = 10.8 = v_Q$, so the particles coalesce and the collision is therefore inelastic.	M1 (AO3.1b) A1 (AO2.1) [2]	Attempt conservation of momentum Find equal velocities and conclude	Do not allow use of KE	
			Total	12			



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Question			Answer/Indicative content	Marks	Part marks and guidance		
4	a		$ R = k v \Rightarrow 2 = 5k \Rightarrow k = 0.4$ $F = ma \Rightarrow f - 0.4v = 4.5 \frac{dv}{dt} \Rightarrow \frac{dv}{dt} = \frac{10f - 4v}{45}$	M1(AO3.3) A1(AO3.3) [2]	AG	Must be clear equation has come from NII	
					<u>Examiner's Comments</u> Almost all candidates could demonstrate how to get to the equation with few dropping 1 or 2 marks for finding the wrong k , or by omitting steps in this 'show that' question.		
	b	i	$\int \frac{1}{10f - 4v} dv = \int \frac{1}{45} dt$ oe $-\frac{1}{4} \ln(10f - 4v) = \frac{t}{45} + c$ $10f - 4v = Ae^{-\frac{4}{45}t}$ $10f - 4u = A$ $4v = 10f - Ae^{-\frac{4}{45}t} \Rightarrow v = \frac{1}{2} \left(5f - (5f - 2u)e^{-\frac{4}{45}t} \right)$ Alternate Method $IF = e^{\int \frac{4}{45} dt} = e^{\frac{4}{45}t}$ $\frac{d}{dt} \left(ve^{\frac{4}{45}t} \right) = \frac{2}{9} f e^{\frac{4}{45}t}$ oe	B1(AO1.1) M1(AO1.1) M1(AO1.1) M1(AO3.4) A1(AO1.1) [5] B1	Correctly separating variables Attempt to integrate - must be in the form $k \ln(a + bv) = pt(+c)$ Exponentiating correctly with a use of law indices / logs so; allow disappearing mod signs Substituting $t = 0$ and $v = u$ into their integrated equation (perhaps manipulated) to find A	Condone missing integral signs $O -\frac{1}{4} \ln(10f - 4u) = c$ r ie substituting $t = 0$ and $v = u$ into their integrated equation (perhaps manipulated) to find c $\frac{10f - 4v}{10f - 4u} = e^{-\frac{4}{45}t}$ ie exponentiating correctly with a use of a law of indices / logs so	



Question			Answer/Indicative content	Marks	Part marks and guidance		
			$ve^{\frac{4}{45}t} = \frac{5}{2}fe^{\frac{4}{45}t} + c$	M1	AG	www	
				M1			
			$c = u - \frac{5}{2}f$	M1			
			$v = \frac{5}{2}f + \left(u - \frac{5}{2}fe^{\frac{4}{45}t}\right) \rightarrow$	A1	Must include e^{kt}		
			$v = \frac{1}{2}(5f - (5f - 2u)e^{\frac{4}{45}t})$	[5]	<u>Examiner's Comments</u> Candidates demonstrated a good understanding of pure mathematics with most candidates being given full marks. The majority of solved this by separating the variables (with only a few unable to do this) with the alternative method being to use an integrating factor. Some candidates lost the accuracy mark for making 'slips' in their workings and recovering from this. As it is a 'show that' question, all workings are taken into consideration to get full marks. Common errors were mistakes with rearranging or exponentiating and care needs to be taken to treat constants appropriately.		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$u < 2.5f \Rightarrow v$ increases (from u) and approaches $2.5f$ as $t \rightarrow \infty$ $u = 2.5f \Rightarrow v = 2.5f$, constant $u > 2.5f \Rightarrow v$ decreases (from u) and approaches $2.5f$ as $t \rightarrow \infty$	B1(AO3.4) B1(AO3.4) B1(AO3.4) [3]	Allow the idea that $v = 2.5f$ for large t , and allow technically inaccurate statements (eg “ v speeds up”) provided that intent is clear See above	SC: If B0B1B0 or B0B0B0 awarded. If mentions v approaches $2.5f$ for cases 1 and 3 award B1 or if mentions v increases in case 1 and v decreases in case 3 award B1
			Examiner's Comments Most candidates were able to identify that when $u = 2.5f$ then the speed is constant. Some candidates identified that there was a limit for cases 1 and 3 but omitted if speed was increasing or decreasing. Higher ability candidates supported their reasoning with a sketch.			

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Question		Answer/Indicative content	Marks	Part marks and guidance		
	c	$\frac{dx}{dt} = \frac{1}{2} \left(5 - e^{-\frac{4}{45}t} \right) f \quad \text{oe}$ $x = 2.5ft + \frac{45}{8} f e^{-\frac{4}{45}t} + c'$ $\Rightarrow c' = -\frac{45}{8} f \text{ and use of } t = 9 \text{ to find } x$ $x = \frac{45}{8} (3 + e^{-0.8}) f$	B1(AO1.1) *M1(AO3.4) Dep *M1(AO1.1) A1(AO1.1) [4]	Could be in a definite integral Substituting $t = 0, x = 0$ to obtain , $x = 2.5ft + \frac{45}{8} f e^{-\frac{4}{45}t}$ with correct use of limits		
		Total	14			

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Question			Answer/Indicative content	Marks	Part marks and guidance	
5		a	$F = k\sqrt{9+1.25^2} = 13$ $k = 4$ $4\sqrt{9+v^2} = 8 \frac{dv}{dt} \Rightarrow \frac{1}{\sqrt{9+v^2}} \frac{dv}{dt} = \frac{1}{2} \quad (\text{AG})$	M1 (AO 3.3) A1 (AO 1.1) A1 (AO 3.3) [3]	Substituting $v = 1.25$ and $F = 13$ Clear use of $F = ma$ leading to AG	
		b	$\int \frac{1}{\sqrt{9+v^2}} dv = \frac{1}{2}t + c$ $\sinh^{-1}\left(\frac{v}{3}\right) = \frac{1}{2}t + c$ $c = 0$ $v = 3\sinh\left(\frac{1}{2}t\right)$	M1 (AO 3.4) M1 (AO 3.4) B1 (AO 1.1) A1 (AO 3.4) [4]	Separating the variables and correctly integrating to obtain bt Integrating other side to obtain $p \sinh^{-1} qv$ Substituting $t = 0, v = 0$ into a solution of the DE to obtain c	Condone omission of c for these two M marks Or using 0 as limits on integrals
		c	$\frac{dx}{dt} = 3\sinh\left(\frac{1}{2}t\right) \Rightarrow x = C + 6\cosh\left(\frac{1}{2}t\right)$ $t = 0, x = 0 \Rightarrow C = -6 \Rightarrow x = 6\left(\cosh\left(\frac{1}{2}t\right) - 1\right)$ oe	M1 (AO 3.4) A1 (AO 2.2a) [2]	Replacing v with $\frac{dx}{dt}$ and integrating to obtain $r \cosh st$	Condone omission of C (or use of 'c' again)
Total				9		



Question			Answer/Indicative content	Marks	Part marks and guidance		
6		a	$T - kv^2 = mv \frac{dv}{dx}$ $m \int \frac{-2kv}{T - kv^2} dv = \int -2k dx$ $m \ln(T - kv^2) = -2kx + c$ $\ln(T - kv^2) = \frac{-2kx + c}{m} \Rightarrow T - kv^2 = e^{\frac{-2kx + c}{m}}$ $\Rightarrow T - kv^2 = e^{\frac{-2kx}{m}} e^{\frac{c}{m}} \Rightarrow v^2 = \frac{1}{k} \left(T - A e^{\frac{-2kx}{m}} \right)$	M1 (AO 3.3) M1 (AO 3.4) M1 (AO 1.1) M1 (AO 1.1) A1 (AO 2.1) [5]	NII with 3 terms and $a = v \frac{dv}{dx}$ used Separating the variables Correctly integrating both sides Exponentiation, giving $f(v) = e^{g(x)}$ AG so intermediate step needed	Condone sign errors and/or missing m oe, eg stating $A = e^{\frac{c}{m}}$	



Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	Work done by constant force is $T(1 - 0) = T$ $x = 0, v = 0$ gives $v^2 = \frac{T}{k} \left(1 - e^{-\frac{2kx}{m}} \right)$ KE gain is $\frac{1}{2}mv_1^2 = \frac{mT}{2k} \left(1 - e^{-\frac{2k}{m}} \right)$ So work done against resistance $= T - \frac{mT}{2k} \left(1 - e^{-\frac{2k}{m}} \right)$ Alternative solution $x = 0, v = 0$ gives $v^2 = \frac{T}{k} \left(1 - e^{-\frac{2kx}{m}} \right)$ WD against resistance $= \int_0^1 kv^2 dx = T \int_0^1 \left(1 - e^{-\frac{2kx}{m}} \right) dx$ $= T \left[x + \frac{m}{2k} e^{-\frac{2kx}{m}} \right]_0^1$ $= T \left(1 + \frac{m}{2k} e^{-\frac{2k}{m}} - \frac{m}{2k} \right)$	B1 (AO 1.1) M1 (AO 3.4) M1 (AO 3.4) A1 (AO 3.4) M1 M1 M1 A1 [4]	Use of initial conditions to find A oe Use of initial conditions to find A oe		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	The limiting velocity is $\sqrt{\frac{T}{k}}$	B1 (AO 3.2a) [1]		
			Total	10		



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Question			Answer/Indicative content	Marks	Part marks and guidance		
7		a	$F = \frac{6}{t^2} - \frac{3v}{t} = 1.5a$ $= 1.5 \frac{dv}{dt} \Rightarrow \frac{dv}{dt} + \frac{2v}{t} = \frac{4}{t^2}$	B1(AO3.3) [1]	AG Using Newton II with $a = \frac{dv}{dt}$ n d rearranging	Initial $F = ma$ equation must be clear	
		b	Integrating $e^{\int \frac{2}{t} dt} = e^{2 \ln t}$ factor is $= t^2$ DE is $\frac{d}{dt}(vt^2) = 4$ $vt^2 = 4t + c$ $v = 0$ when $t =$ gives $0 = 8 + c$ $vt^2 = 4t - 8$ when $t = 20$, speed of the piston is 0.18 ms^{-1}	M1(AO1.1a) A1(AO1.1) M1(AO3.4) A1(AO1.1) M1(AO3.4) A1(AO1.1) A1(AO2.2a) [7]	Using their integrating factor Use of $v = 0$, $t = 2$ in their equation to find a constant of integration		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	$\frac{dx}{dt} = \frac{4}{t} - \frac{8}{t^2} \Rightarrow \text{distance} = \int_2^{20} \left(\frac{4}{t} - \frac{8}{t^2} \right) dt$ $= 5.6103... \text{ so distance is } 5.61 \text{ m correct to 3sf}$ Alternative solution $\frac{dx}{dt} = \frac{4}{t} - \frac{8}{t^2} \Rightarrow x = 4 \ln t + \frac{8}{t} + C$ $x = 0 \text{ when } t = 2 \text{ gives } C = -4 \ln 2 - 4$ $\text{so } t = 20 \text{ gives } x = 4 \ln 20 + 0.4 - 4 \ln 2 - 4 = 5.61 \text{ (3sf)}$	M1(AO3.4) A1(AO3.4) A1(AO1.1) M1 A1 A1 [3]	$Us^v = \frac{dx}{dt} a$ e n d expression of distance moved as a definite integral Correct integrand and limits BC; AG so justification for 3sf accuracy is needed $Us^v = \frac{dx}{dt} a$ e n d attempt to integrate AG	(C = -6.77...) (5.61034...)
			Total	11		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
					others showed this when they needed to replace $\sin \theta$ by θ in their proof. Some candidates only realised the need to show this when they needed to find the amplitude in part (ii) and, in this case, credit was awarded in part (i) for the work shown in part (ii).	
		ii	$0.087266 = A \sin 3.5t$ $t = 0.238 \text{ secs}$ $t' = 2 \left(\frac{1.7952}{4} - 0.2378 \right) = 0.422(\text{s})$ $\dot{\theta} = 0.118 \times 3.5 \cos 3.5 \times 0.238$ Linear speed = $0.222(\text{ms}^{-1})$	M1 A1 A1 M1 A1 [5]	OR $5 = A \sin 3.5t$; $A =$ amplitude Or 0.65972 Or $0.65972 - 0.2378$ OR $\dot{\theta} = \sqrt{(3.5^2 (0.118^2 - 0.0873^2))} \times 0.278$ May use $\cos^*$ 0.2378 allow $\sin$ if consistent with $*$; allow 5° and 6.76° Or $\frac{1}{2} m \frac{9}{90} = \frac{1}{2} m v^2 + mg 0.8(1 - \cos 5^\circ)$	Examiner's Comments Candidates with the correct amplitude were usually able to find the times taken for the string to make an angle of 5° with the vertical, though not all found the time that elapsed between the two occasions. Candidates with the wrong amplitude were allowed partial credit for a correct method to find a time, and for a correct method to find the linear speed of P .



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		iii	Max height is still 0.00556 so i $\text{Max angle} = \cos^{-1} \frac{(0.05 - 0.00556)}{0.05} [27.3]$ Not SHM since angle is not small	B1 B1 B1 [3]	accept 'still the same' or attempt to work out height 0.476 rads	
					Examiner's Comments A small number of candidates answered this very well, by finding the angle made with the vertical by the string when <i>P</i> at its highest point to the right of <i>Q</i> and explaining that this angle is too big for the approximation of $\sin \theta$ by θ to be valid. Many candidates gained partial credit for an answer stating that the angle was too big without showing any detailed working. Other candidates gave wrong answers referring to the string becoming slack, <i>P</i> moving in a circle or other forces being involved to suggest that the motion is not simple harmonic.	
			Total	15		

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Question			Answer/Indicative content	Marks	Part marks and guidance		
9		i	48 (ms ⁻¹)	B1 [1]	Accept ≤ 48 Examiner's Comments This was usually correct, though 12 (m s ⁻¹) was a common wrong answer.		
		ii	Use $\frac{1}{2}\sqrt{12 - \frac{1}{4}v} = 0.2a$ $\frac{1}{2}\sqrt{12 - \frac{1}{4}v} = 0.2 \frac{dv}{dt}$ $2.5t = \int \frac{dv}{\sqrt{12 - \frac{1}{4}v}} (+c)$ $2.5t = -8\left(12 - \frac{1}{4}v\right)^{\frac{1}{2}} (+c)$ [c = 24] $v = 48 - 4\left(3 - \frac{t}{3.2}\right)^2$	M1* A1 *M1* A1 *M1 A1 [6]	Accept $v \frac{dv}{dx}$ for a Sep variables and integrate one side For attempt to find c, dep previous M1 $12 + 7.5t - \frac{25}{64}t^2$ (0.390625)	Allow missing 0.2 or sign error $2.5t = \int \frac{2dv}{\sqrt{48-v}} (+c)$ $2.5t = -4(48 - v)^{\frac{1}{2}} (+c)$ $v = 48 - 0.390625(9.6 - t)^2$	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
					difficulty converting their expression for t to an expression for v . Some candidates gave their answer as a quadratic expression in t ; any correct answer in the form $v = 48 - A\left(B - \frac{t}{C}\right)^2$ was also accepted.	
		iii	$x = \int \left(12 + \frac{24}{3.2}t - \frac{4}{3.2^2}t^2\right) dt$ $x = 12t + 3.75t^2 - 0.1302t^3 (+c)$ $(t = 0 \text{ and } t = 3.2)$ Distance = 72.533 (m)	M1 A1 M1 A1 [4]	OR $x = \int \left(48 - 4\left(3 - \frac{t}{3.2}\right)^2\right) dt$ $x = 48t + \frac{12.8}{3}\left(3 - \frac{t}{3.2}\right)^3 (+c')$ via subst $x = 12.8\left(12u - \frac{u^3}{3}\right) + c$ OR $\frac{1}{2}\sqrt{12 - \frac{1}{4}v} = 0.2v \frac{dv}{dx}$ ft their (ii)	
					Examiner's Comments Candidates with the wrong answer to part (i) were often able to gain a mark for writing their v as $\frac{dx}{dt}$ and another for showing the need to convert the limits for v given in the question to limits for x . The most common error on this part by candidates using a correct expression for v from part (ii) was to use the given limits of 12 and 32 as limits for t .	
			Total	11		



Question			Answer/Indicative content	Marks	Part marks and guidance		
10		a	(i) $m \frac{dv}{dt} = -mg - (-6mv + mv^2)$ $\frac{dv}{dt} = -\frac{1}{v^2 - 6v + 9.8}$ $= -\frac{1}{(v-3)^2 - 9 + 9.8} = -\frac{1}{(v-3)^2 + 0.8}$	B1(AO3.3) M1(AO1.1) E1(AO2.1) [3]	Divide by m and rearrange AG Completing the square in the denominator		
		a	(ii) $t = -\int \frac{dv}{(v-3)^2 + 0.8}$ $t_1 = -\int_{8.25}^0 \frac{dv}{(v-3)^2 + 0.8}$, and $t_2 = -\int_{8.75}^0 \frac{dv}{(v-3)^2 + 0.8}$ $t_1 = 2.9997903...$ $t_2 = 3.0159247...$ $2.9997... < t < 3.0159...$ so B 's model for the vast majority of speeds in the interval $(8.5 \pm 0.25) \text{ m s}^{-1}$ does produce a value of t greater than 3 so it is extremely likely (but not certain) that B 's model is consistent with the time recorded by A .	M1(AO3.1a) B1(AO3.4) A1(AO1.1) BC A1(AO1.1) BC E1(AO2.2b) [5]	Separation of variables Using the limits correctly at least once Or use $\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$ BC BC Indication that the non-exact nature of the initial speed leads to a result that is likely to be correct		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		b	e.g. student C's claim is incorrect as they have assumed that the differential equation for the motion of the particle is the same in both directions In fact the differential equation for when the particle is falling is given by $mg - (-6mv + mv^2) = m \frac{dv}{dt}$ $-\int_0^1 \frac{1}{(v-3)^2 - 18.8} dv$	B1(AO2.2a) B1(AO2.3) B1(AO3.3) [3]	For explaining that the differential equation of motion is not the same in each direction For either this corrected model or stating that the two forces in the vertical direction are now acting in opposite directions	
			Total	11		



Question			Answer/Indicative content	Marks	Part marks and guidance		
11		a	$\mathbf{a} = \frac{d\mathbf{v}}{dt} = 2t\mathbf{i} - \frac{10}{(2t+1)^2}\mathbf{j}$ $\mathbf{F} = m\mathbf{a} = 4t\mathbf{i} - \frac{20}{(2t+1)^2}\mathbf{j}$	M1(AO1.1) A1(AO1.1) A1FT(AO3.3) [3]	Differentiating one term of $\mathbf{v}$ correctly FT their $\mathbf{a}$		
		b	$\mathbf{F} \cdot \mathbf{v} = 4t(t^2 - 3) - \frac{100}{(2t+1)^3}$ Power is 208 W	M1(AO3.4) A1(AO1.1) A1(AO1.1) [3]	Attempt at scalar product 207.86282 ...	OR M1 at $t = 4$ $-(10\mathbf{i} - \frac{10}{8}\mathbf{j}) \cdot (13\mathbf{i} + \frac{5}{9}\mathbf{j})$	
		c	$T = \frac{1}{2}m\mathbf{v} \cdot \mathbf{v} = (t^2 - 3)^2 + \frac{25}{(2t+1)^2}$ Work done is 167 J	B1(AO3.4) M1(AO1.1) A1(AO1.1) [3]	$T(4) - T(2)$ 167.30864 2...	OR B1 for value of T at 2 or 4 M1 $T(4) - T(2)$	
			Total	9			



Question			Answer/Indicative content	Marks	Part marks and guidance	
12		i	$3\cos 2t = 0.2 \frac{dv}{dt}$	M1*	Use of $F = ma$; condone wrong / missing 0.2 and wrong sign	Examiner's Comments Most candidates were able to use Newton's second law correctly and to integrate the equation they found. A small number of candidates omitted to give the expression for the velocity explicitly. Rather more candidates had trouble with the minimum value of the velocity, with 4 m s^{-1} and 0 m s^{-1} being seen on some scripts.
		i	$7.5\sin 2t = v(+c) \text{ oe}$	*M1	Attempt to integrate, one side correct; condone missing c ,	
		i	$v = 7.5 \sin 2t + 4$	A1	CAO	
		i	$11.5 \text{ (ms}^{-1}\text{) and } -3.5 \text{ (ms}^{-1}\text{)}$	A1	Depends on both M marks and fully correct working	
		ii	$x = -\frac{15}{4}$	M1*	Ft if (i) has sin or cos term	No need to find c (3.75)
		ii	Ave speed is their distance/ $\frac{\pi}{2}$	*M1		
		ii	$\left(\frac{15}{4} + 6\pi\right) - \left(-\frac{15}{4} + 4\pi\right)$	*M1	$\{x(\frac{3}{2}\pi) - x(\pi)\}_{\text{found;}}$	$\left(\frac{15}{2} + 6\pi\right) - (+4\pi)$



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	8.77 (ms ⁻¹)	A1	CAO Accept $\frac{15}{\pi}$	Examiner's Comments Most candidates did this well. Those with a small error in velocity in part (i) were able to gain some credit. A small number of candidates did not understand that they had to find the distance travelled in order to find the average velocity.
			Total	8		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
13		i	$-\frac{v}{8} = 0.4 \frac{dv}{dt}$	M1*	Allow sign error, allow 0.4a	
		i		A1		
		i	$t = -3.2 \int \frac{1}{v} dv$	*M1	Attempt to separate variables and integrate	Or $t = -3.2 \int_{10}^5 \frac{1}{v} dv$
		i	$t = -3.2 \ln v + 3.2 \ln 10$	A1		$t = -3.2 \ln v$; limits used correctly
		i	Time taken = 3.2ln2 or 2.22 (s)	A1	Examiner's Comments A small number of candidates made errors with this part, including an error with the minus sign, using $a = v \frac{dv}{dx}$ and using limits the wrong way round.	2.21807...
		ii	$-\frac{v}{8} = 0.4v \frac{dv}{dx}$	M1*	Allow sign error	
		ii		A1		
		ii	$x = -3.2 \int dv$	*M1	Attempt to separate variables and integrate	
		ii	$x = -3.2 v + 32$	A1	$x = 16$ when $v = 5$.	
		ii	ave speed = $x/(i)$	*M1		Their x evaluated
		ii	ave speed = 7.21	A1		Accept 5/ln2
		ii	OR $\frac{dx}{dt} = 10e^{-\frac{t}{3.2}}$	M1*	For M1, ft from (i), must contain ln term	
		ii		A1		
		ii	$x = 10 \int e^{-\frac{t}{3.2}} dt$	*M1	Attempt to separate variables and integrate	
		ii	$x = 32 \left(1 - e^{-\frac{t}{3.2}}\right)$	A1	Must show constant or use limits correctly	$x = 16$ when $t = 3.2 \ln(2)$
		ii	ave speed = $x/(i)$	*M1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	ave speed = 7.21	A1	Dep all 5 previous marks Examiner's Comments About half the candidates used the equation they had found in part (i), and wrote it as $\frac{dy}{dx} = 10e^{-\frac{t}{2.2}}$, though some took a long time to get to this stage. $a = v \frac{dv}{dx}$, and had a rather more straightforward calculation to do. In both cases, a small number of candidates made sign errors, or omitted the arbitrary constant of integration.	Accept 5/ln2
			Total	11		



Question			Answer/Indicative content	Marks	Part marks and guidance	
14		i	Use of $F = ma$, using $v \frac{dv}{dx}$	M1*	Examiner's Comments Most candidates correctly used $v \frac{dv}{dx}$ for acceleration, with a very small number making a sign error, missing the mass of 0.3 kg or using equations of motion for constant acceleration. Almost all of those who had the correct start went on to correctly establish the given expression. Most candidates established that the arbitrary constant was 0, though not doing this was not penalised	Allow sign error / 0.3 omitted
		i	$0.3v \frac{dv}{dx} = 1.5x$	A1		
		i	Attempt to rearrange and integrate	*M1	$0.3v^2 = 1.5x^2 (+c)$	No need for c. At least one side integrated correctly
		i	$v = \sqrt{5x}$ AG	A1	correct derivation WWW	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Integrate to find x in terms of t	M1	$dx/x = \sqrt{5}dt$ and int 1 side correctly Examiner's Comments some candidates worked from the given answer by expressing v as $\frac{dx}{dt}$, others started with $\frac{dv}{dt} = 5x$ and, provided they managed to write $5x$ in terms of v they made progress; however, some candidates attempted to solve the differential equation written in terms of the 3 variables t, v and x . In this part candidates were required to either find the arbitrary constant, or show that it was 0 – depending on their method of solution. A number of candidates made mistakes involving the $\sqrt{5}$ factor, either losing it or putting it in the wrong place in an equation.	Need to separate variables
		ii	$\ln x = \sqrt{5}t + c$	A1		No need for c for first 2 marks
		ii	$x = e^{\sqrt{5}t}$	A1		Must include showing $c = 0$.
		ii	$v = \sqrt{5}t e^{\sqrt{5}t}$	A1	CAO	
		ii	OR Integrate to find v in terms of t	M1	Use in $0.3v \frac{dv}{dx} = 1.5x$ and int 1 side correctly	No need for c for first 2 marks
		ii	$\frac{dv}{v} = \sqrt{5}dt$			
		ii	$\ln v = \sqrt{5}t + c$	A1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\ln v = \sqrt{5} + \ln(\sqrt{5}t)$	A1	Must include showing $c = \ln(\sqrt{5})$	
		ii	$v = \sqrt{5} e^{\sqrt{5}t}$	A1	CAO	
			Total	8		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
15		i	$a = 0.6 \text{ (m)}$	B1	Enter text here.	
		i	$\omega = 4$	B1	Accept sight of $\frac{\pi}{0.25\pi}$ or $\frac{2\pi}{0.5\pi}$	
		i	Max vel = $a\omega = 2.4 \text{ (m s}^{-1}\text{)}$	B1ft	ft from wrong a and / or ω Examiner's Comments Although most candidates found this introductory part straightforward, the amplitude was sometimes given as 1.2 m or 0.3 m and ω as 8 s^{-1} or 2 s^{-1} . Almost all realised that the maximum velocity was given by $a\omega$, though some used $v^2 = \omega^2(a^2 - x^2)$.	
		ii	<i>Must use their a and ω from (i) unless defined differently in (ii)</i> $x = 0.6\cos 4 \times 0.7$	M1	Use of $a\cos\omega t$; complete method	Or $a\sin(\omega t + \epsilon)$, with $\epsilon = \pm \pi/2$
		ii	$x = -0.565$	A1		-0.565333...
		ii	$v = -0.6 \times 4 \times \sin 4 \times 0.7$	M1	Use of $(-)\omega\sin\omega t$ or $v^2 = \omega^2(a^2 - x^2)$	Or $(-)\omega\cos(\omega t + \epsilon)$, with $\epsilon = \pm \pi/2$; Allow M1ft from wrong formula for x

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$v = -0.804$	A1	If v^2 formula used, direction of v needs to be made clear. Examiner's Comments Most candidates used the correct formulae to find the value of x and the velocity, though some used the wrong trigonometric function. A few candidates employed $\omega t + \varepsilon$, though rarely correctly. A small number of candidates found the correct value of x and then went on to add 0.6 m to it. Candidates using $v^2 = \omega^2(a^2 - x^2)$ were expected to make the direction of the velocity clear.	-0.80397.....
		iii	<i>Do not accept answers from wrong working</i> t and x for one point	B2	Values of t are = 0.0854, 0.871	$\pi/4 - 0.7$, $\pi/2 - 0.7$
		iii	t and x for second point	B1	Values of x are 0.565, -0.565	Ignore ref to point when $t = 0.7$
		iii	Correctly giving precisely 2 other	B1	Dep first 3 marks	Can show on diagram

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	Occasions, with x and t values matching sc, if < 3 scored, both t values Or one t value B1 Or $x = 0.565$ B1 Of B0 scored allow B1 for number of other occasions shown to be 2	B2	Ignore wrong values “ ” Examiner's Comments Quite a number of candidates found the values of t from a consideration of the motion of the particle, with a lot of helpful diagrams seen. Any mistakes made in finding a or ω earlier restricted the mark in this part. Some candidates did this part by solving equations such as $v = -a\omega \sin\omega t$; these candidates could only gain marks in this part if they had correct values in earlier parts.	P has this speed 4 times in 1 period (1.570 s) so 2 other times in $0 < t < 1$
			Total	11		



Question			Answer/Indicative content	Marks	Part marks and guidance	
16		i	Use of energy equation	M1	Examiner's Comments This was done well by many candidates.	Allow M1 if sign error and / or 9.8 missing and / or missing m or l
		i	$0.5 \text{ m } (0.3)^2 = m \times 9.8 \times 0.8 \times (1 - \cos \theta)$	A1		
		i	$\theta = 0.107$	A1	No errors AG	0.107194171
		ii	Use $F = ma$	M1	$m \times 9.8 \sin < 9 = -m \times 0.8 \ddot{\theta}$ Examiner's Comments Some candidates missed out the length of the pendulum, and a few others only quoted the SHM result; a few failed to work out the period of the motion.	allow M1 if sign error, or 9.8 missing
		ii	$\ddot{\theta} = -12.25 \theta$	A1		Allow fraction
		ii	small θ	B1	Dep on having seen $\text{acc} = k \sin \theta$	Rigorous
		ii	Use of $T = \frac{2\pi}{\omega}$	M1	or sight of $\omega = 3.5$	
		ii	$T = 1.80$	A1		accept $\frac{4\pi}{7}$ (1.795195)



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	identifying amplitude as 0.107	B1	Examiner's Comments This part proved more challenging, though most candidates did tackle it by means of the equations for velocity and displacement in terms of $\cos \omega t$ and $\sin \omega t$. Those who chose to use $\theta^2 = \omega^2(\alpha^2 - \theta^2)$ were able to find the angular displacement but were not usually able to correctly find the time, since this required some manipulation involving the period of motion. For both methods, some candidates used a linear speed of -0.2 m s^{-1} when they should have used an angular speed of -0.25 rad s^{-1} .	ft from (i)
		iii	Use of $\dot{\theta} = 0.107 \times 3.5 \times \cos(3.5t)$	M1	or $\sin(3.5t + \epsilon)$, ϵ not 0	ft for a and ω ; allow sign error
		iii	Use of $\dot{\theta} = -0.25$	A1	Consistent angle or length	
		iii	$t = 0.658$	A1		(0.6576339)
		iii	Use of $\theta = 0.107 \sin(3.5t)$	M1	ft from velocity equation (matches, ignore sign)	
		iii	$(\theta \Rightarrow) 0.0797 \text{ rads}$	A1	accept 5.20°	(0.0796678 or 0.079576)
			Total	14		



Question			Answer/Indicative content	Marks	Part marks and guidance	
17		i	Use $F = mv \frac{dv}{dx}$	M1		Allow sign error, missing m or g inc
		i	$-4v = \frac{dv}{dx}$	A1	Expression for $\frac{dv}{dx}$ required	
		i	$-4x = \ln v + c$	M1	Get (+/-) $Ax = \ln v + c$	
		i	$0 = \ln 2 + c$ $\ln \frac{v}{2} = -4x$	M1	Valid attempt to find c Need a step leading to given answer	
		i	$v = 2e^{-4x}$	A1	AG	
					Examiner's Comments	
					This question was done well, with most candidates successfully establishing the given answer. There were indications that some candidates had sensibly used the given answer to go back and correct slips in their working. However, quite a number did not give an explicit expression for $\frac{dv}{dx}$ which was asked for in the question. Most dealt competently with the arbitrary constant; those using limits for the integration found the answer rather more efficiently.	
		ii	$e^{4x} dx = 2 dt$	M1*	Write v as $\frac{dx}{dt}$ and separate variables	$dv/4v^2 = -dt$
		ii	$\frac{1}{4} e^{4x} = 2t + c$	A1	Must have c or use limits	$\frac{1}{v} = 4t + \frac{1}{2}$



Question			Answer/Indicative content	Marks	Part marks and guidance	
	ii		$\frac{1}{4} = 0 + c$	*M1	Valid attempt to find c or subst limits	$\frac{dx}{dt} = \frac{2}{8t+1}$ OR $t = 0.5$ gives $v = 0.4$
	ii		$e^{4x} = 4(1 + \frac{1}{4})$	*M1	Find x when $t = 0.5$ - need to remove exp; Allow even if no c	$x = \frac{1}{4} \ln(8t + 1) + c$ OR $-4x = \ln 0.2$
	ii		$x = \frac{1}{4} \ln 5$	A1	Accept 0.402(359...) Examiner's Comments This part proved a little more challenging than part (i), and gave rise to quite a range of methods. Many started with the equation established in part (i), others started with $\frac{dv}{dt} = -4v^2$ and established that $\frac{1}{v} = 4t = \frac{1}{2}$. From here, some integrated again, others substituted $t = 0.5$, while others now used the expression for v in terms of x from part (i). Whichever of the two main methods was used, some candidates went wrong by omitting the arbitrary constant.	$x = \frac{1}{4} \ln 5$
Total				10		



Question			Answer/Indicative content	Marks	Part marks and guidance	
18		i	Use of $F = ma$	M1	$\frac{3}{2}t - 1 = 0.2 \frac{dv}{dt}$ Examiner's Comments Most candidates gained full marks. Some candidates had the force in the opposite direction, some ignored the need for an arbitrary constant after the integration, and a small number left their answer as $0.2v = \dots$	allow sign errors or m omitted
		i	Integrate correctly	A1	$v = \frac{15}{4}t^2 - 5t(+c)$	allow if c missing or wrong
		i	$v = \frac{15}{4}t^2 - 5t + 0.8$	A1		oe
		ii	Use vel = 0.8	M1	$\frac{15}{4}t^2 - 5t + 0.8 = 0.8$ Examiner's Comments Almost all candidates gained M1 for putting their answer to part (i) equal to 0.8. The A1 mark was only awarded to those who gained the correct answer having used a correct equation from part (i).	ft their (i)
		ii	$t = 1.33$ (s) or $1\frac{1}{3}$ (s)	A1	must come from correct equation for v	Accept 4/3
		iii	Integrate to find x	M1*	At least 2 terms with powers increased by 1	
		iii	$x = \frac{15}{12}t^3 - \frac{5}{2}t^2 + 0.8t$	A1	Need to state $c = 0$, or use limits	
		iii	Solve for $x = 0$	*M1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$t = 1.6 \text{ (s) or } 0.4 \text{ (s)}$	A1	Both answers needed; must be from correct work to find equation Examiner's Comments Most candidates integrated their answer to part (i), with just a few losing a mark by failing to establish that the arbitrary constant was 0, and went on to solve a quadratic equation to give the two times required. A small number of candidates went back to the acceleration equation, used $v \frac{dv}{dx}$ for acceleration and failed to do the question correctly.	Ignore $t = 0$
		iv	$x(3) - x(2)$	M1	Allow for $x(2)$ or $x(3)$ worked out from (iii)	13.65 or 1.6
		iv	Distance is 12.05 (m)	A1		Accept 12 or 12.1
			Total	11		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
19		i	Use of $F = ma$ when string stretched	M1	Must have mg – tension term (involving $39.2m$, 0.8 and x) = ma $mg - \frac{39.2m(x - 0.8)}{0.8} = m\ddot{x}$ Examiner's Comments While many candidates did this completely correctly, quite a large number of candidates only established that the particle is in equilibrium at a point 1 m below O , while others only established that the particle performs SHM without any consideration of the centre of motion.	allow if sign errors; x could be length or ext of string, or from eq ^m pos. $mg - \frac{39.2mx}{0.8} = m\ddot{x}$ leads to
		i		A1	$\ddot{x} = -49(x - 1)$	$\ddot{x} = -49(x - 0.2)$ $mg - \frac{39.2(x + 0.2)}{0.8} = m\ddot{x}$ leads to $\ddot{x} = -49x$
		i	Show $x = 1$ is centre of SHM or that $x = 1$ is equilibrium position.	B1	and state about $x = 1$	Convincingly
		ii	By energy	M1	Must be PE term and EE term Examiner's Comments This was generally done successfully, with candidates either working in terms of the extension of the string or the distance below of P below O .	Allow for missing '2', wrong 'g' or inconsistent lengths



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		A1	$mg(0.8 + e) = \frac{39.2me^2}{2 \times 0.8}$	Or $mgh = \frac{39.2m(h - 0.8)^2}{2 \times 0.8}$
		ii	$e = 0.8$ satisfies this equation AG	A1	Or by solving quadratic in e Allow full credit if done correctly from $v^2 = \omega^2 (a^2 - x^2)$	and $h = 0.8 + e$ $2.5e^2 - e - 0.8 = 0$ Convincingly Allow integration of $v \frac{dv}{dx} = g - 49x$



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	For SHM, $\omega = 7$	B1		To be awarded if seen in (i) or (iv) Examiner's Comments Almost all candidates found this part difficult, with only a few succeeding completely. Most candidates realised that the motion was partly free fall and partly under SHM. However, not all candidates realised that the free fall was for a distance of 0.8 m, and quite a number failed to make any use of a standard SHM equation for displacement in terms of time. Correct solutions were seen based on $x = 0.6\cos 7t$ and $x = 0.6\sin 7t$ and, less often, $x = 0.6\cos(7t + \epsilon)$ and $x = 0.6\sin(7t + \epsilon)$. However, using a correct formula does not guarantee success – the candidate has to use it for the right portion of the motion. In this question starting at $x = -0.2$ caused many candidate to make errors. Very few candidates used any form of diagram of what was happening, which could have made it easier for them to be successful.
		iii	$a = 0.6$	B1		or seen or used here
		iii	Correct use of appropriate SHM distance equation	M1	$-0.2 = 0.6 \cos(7t)$ or $-0.2 = 0.6 \sin(7t)$	Allow +0.2, allow their a and ω
		iii	$t = 0.272$ (9476) from bottom ($x = 1.6$) to $x = 0.8$	A1	Could be $0.0485 + 0.224$	
		iii	$t = 0.404$ (061) from O to $x = 0.8$	B1	Or $\frac{2\sqrt{2}}{7}$	May be seen first

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	Time to reach lowest point = 0.677 s	A1ft	('0.273' + '0.404')	
		iv	Use of $v = -a\omega\sin\omega t$ or $a\omega\cos\omega t$	M1	Must ft from their 'x' equation in (iii), or shown here Examiner's Comments Most candidates realised that they needed to use a standard SHM equation here, but many had difficulty finding the correct time to use with it. This time, of course, depended on the approach they used, which had either been decided in part (iii), or was shown in this part by a fresh start.	Allow use of their a and ω , sign error
		iv	$v = -0.6 \times 7\sin 7t$	A1	or $0.6 \times 7\cos 7t$	
		iv	Use of $t = 0.8 - 0.677 = 0.123$ after bottom point	B1ft	Or use of $t = 0.3475$ in 'cos' version	Must be between 0 and 0.8
		iv	$v = 3.19$ (3.185677...)	A1	(-)3.187	Do not allow if direction stated to be down.
			Total	16		

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